

Capstone Part 1 Overview: Define and Solve an ML Problem Using a Traditional Model

In this lab, you will begin a capstone project in which you will follow the steps of the machine learning life cycle to solve a binary classification problem using a traditional machine learning model and a neural network, and compare their performance.

This capstone spans **two lab sessions**.

Unit 5 Lab: You will define the problem, explore and prepare the data, and train, test, evaluate, and improve one of the supervised learning models you have worked with so far in this course.

Unit 6 Lab: After completing Unit 6 on neural networks, you will implement a neural network model to solve the same problem and compare the performance of the two models.

You will choose a machine learning problem based on one of the two data sets you have worked with in this course. Below is a business brief for each data set. Read through each business brief. Once you choose which data set you would like to work on across both labs, you may want to revisit the associated brief below.

▼ Business Brief A: EquiLend

Dataset: Census Income data set

You will be putting yourself in the shoes of a junior machine learning engineer (MLE) at EquiLend, a company tasked with developing a machine learning model to predict individual income levels for one of its clients.

1. Company and Context

EquiLend is a financial technology company based in Chicago that builds predictive models for lending institutions, credit unions, and nonprofit financial services organizations. Its models help clients make faster, more consistent decisions about financial product eligibility by replacing or supplementing manual review processes with data-driven predictions.

2. Business Challenge

EquiLend's new client is a nonprofit community development financial institution (CDFI) that provides low-interest loans and financial education programs to underserved communities across the Midwest. The CDFI currently determines program eligibility through a manual intake process in which case managers review

each applicant's employment status, education level, hours worked, and other background information before making a recommendation. This process is time-consuming, inconsistently applied, and difficult to scale as the CDFI expands its reach.

The CDFI has provided EquiLend with a dataset based on U.S. Census demographic and employment records. The client wants to explore whether a machine learning model can reliably identify individuals likely to be earning above or below a key income threshold — information central to determining eligibility for several of its programs.

3. Business Goal

EquiLend's goal is to build a binary classification model that predicts whether an individual's annual income exceeds \$50,000. The model should learn from employment features in the Census dataset and generalize reliably to new applicant data. If successful, it could help the CDFI triage applications more efficiently and target outreach efforts toward individuals who may qualify for its financial programs.

4. Your Role and Task

You have just joined EquiLend as a junior MLE on the eligibility modeling team. The team has been given the Census dataset and tasked with building a machine learning model that will predict whether an individual's annual income exceeds \$50,000. Your job is to complete the full machine learning life cycle to implement a binary classifier.

Building a machine learning model involves making decisions throughout the machine learning life cycle. Choices made during data preparation — such as handling missing values, selecting features, and addressing class imbalance — affect the quality of the training data. Choices made during model development and evaluation — such as selecting a model, tuning hyperparameters, and evaluating performance — affect how well the model generalizes to new applicant data.

One dimension of this work that deserves particular care is fairness. The Census dataset includes demographic features such as sex, race, and native country. A model trained on this data could encode historical inequalities rather than correct for them. As you build your model, you should think critically about which features you include, what your model's errors look like across different groups, and what the consequences of a wrong prediction might be for the individuals affected.

Business Brief B: Trove Analytics